

Foundations of Machine Learning – Final Group Project

Deadline

Friday, August 29 at 17:00

What to Submit

Each group must submit:

- A Jupyter Notebook (.ipynb) that:
 - Contains all the code used in the project.
 - Runs start to finish without errors.
 - Includes markdown cells that explain your reasoning and results.
- A PDF report (.pdf), written in one of three styles:
 - A Scientific Report (Track 1, see below)
 - Or a Technical Report (Track 2, see below)
 - Or a GPT Report (Track 3, see below)

Project Goal

Design and implement a simple feedforward neural network to solve a regression or classification problem (or, for Track 3, a small GPT-style language model).

You are allowed to use only:

- Python standard libraries
- NumPy
- Matplotlib, Seaborn, and scikit-learn (for visualization or preprocessing only)

Note: Track 1 (Scientific Report) is exempt from this restriction. Since the focus of that track is experimental analysis rather than implementation, you may use an existing, already-written network (e.g., from PyTorch, TensorFlow, or scikit-learn's MLP) instead of building one from scratch.

Three Project Tracks

Track 1: Scientific Report

- You explore a new or interesting dataset (ideally a small or underexplored one).
- You only need to implement (or use an existing) one or two simple networks (e.g., basic MLPs).

- You focus your report on experimental outcomes, data-driven insights, or hypothesis testing.
- The report should follow a standard scientific format: Introduction, Methods, Results, Discussion.
- *You may use an already-written/pre-built network implementation (e.g., from PyTorch or scikit-learn) rather than implementing one from scratch — see Requirements below.*

Track 2: Technical Report

- You choose a standard dataset or task.
- You start with a basic neural network (e.g., a single hidden layer) and incrementally improve it.
- You may explore deeper MLPs, convolutional networks, or attention-based layers (implemented manually).
- You perform hyperparameter searches, ablation studies, and diagnostics.

Track 3: GPT Report

- You implement a small GPT-style (decoder-only Transformer) language model from scratch using NumPy.
- You do **not** need to implement byte-pair encoding (BPE) yourself — only the GPT architecture itself needs to be built from scratch. You may use an existing tokenizer (e.g., a simple character-level tokenizer, or a library such as tiktoken) for preprocessing.
- Andrej Karpathy's "[Let's build GPT from scratch](#)" (and the accompanying [nanoGPT repository](#)) is a good reference for the expected scope and level of implementation detail.
- You train your model on a small text corpus and demonstrate training progress and text generation.
- Given the added complexity of this track, a small dataset and modest model size (e.g., a few layers, a small context window) are expected and appropriate.

Requirements for All Projects

- Implement your neural network from scratch using NumPy
- Use the network for regression, classification, or (for Track 3) next-token language modeling.
- Use a real-world dataset with at least moderate complexity.
- Demonstrate training, evaluation, and interpretation of your model's results.
- Provide insight into hyperparameter choices and learning behavior.

Examples of Good Projects

Computer Vision

- Image classification on CIFAR-10 using MLP and hand-crafted CNN

Track: Technical

- Facial expression recognition using basic MLP or CNN

Track: Scientific or Technical

Language

- Sentiment classification on IMDB reviews using bag-of-words

Track: Technical

- Character-level GPT trained on a small corpus (e.g., Tiny Shakespeare) to generate text in a given style

Track: GPT

Examples of Projects to Avoid

- Using datasets that are too large (e.g., ImageNet)
- Using only scikit-learn classifiers (you must implement or use a neural network, not a non-neural classifier)
- Submitting notebooks that cannot run or have incomplete results

Suggested Dataset Sources

- <https://archive.ics.uci.edu/>
- <https://www.kaggle.com/datasets>
- <https://openneuro.org/>
- <https://physionet.org/>
- <https://huggingface.co/datasets>

Report Formats

Scientific Report (~6 pages max)

- Introduction and Research Question
- Dataset Description
- Methods (model setup, training)
- Results and Analysis
- Discussion and Interpretation
- References

Technical Report (~6 pages max)

- Problem Description
- Dataset Overview

- Baseline Neural Network
- Iterative Improvements
- Evaluation (e.g., accuracy plots, comparison)
- Lessons Learned and Challenges

GPT Report (~6 pages max)

- Problem Description and Motivation
- Dataset and Tokenization Approach
- GPT Architecture (design choices, model size)
- Training Process
- Generation and Evaluation (loss curves, sample generations, qualitative assessment)
- Lessons Learned and Challenges

Questions?

If you have any questions or are unsure about your topic or track, please ask during office hours or reach out by email.